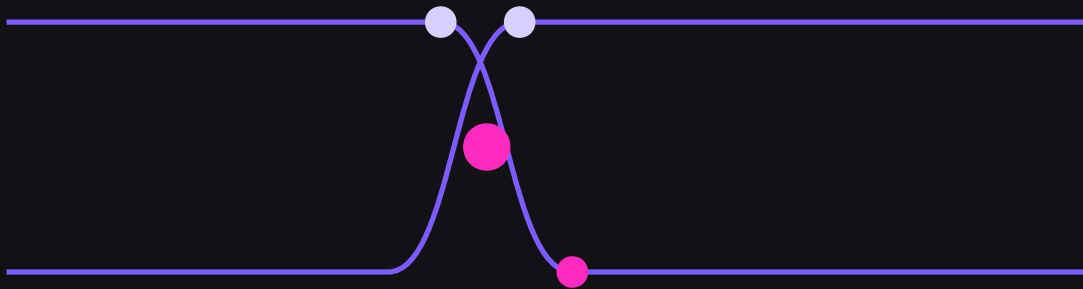


Where the Machine Ends

What changes when the components of a robot are alive?

A designed article on robotics, embodiment and the changing boundary between engineered and living systems.



What changes when the components of a robot are alive?

Engineering has spent centuries trying to make components boring.

Steel doesn't mature. A bearing doesn't need oxygen. A motor doesn't reorganise itself. Predictable is the whole point.

Then, in 2025, researchers in Japan built a robotic hand powered by cultured human muscle. It had articulated fingers, tendon-like cables and an artificial skeleton, and five bundles of engineered skeletal muscle pulled the fingers into motion.^[1]

Living muscle develops. It needs nutrients. Its performance changes with use. It tires, recovers, gets injured and eventually dies. A familiar robotic component arrives with a physiology.

The component that needs to eat

Biohybrid robotics already includes walkers, swimmers, grippers and other machines powered by skeletal or cardiac muscle, sometimes combined with neurons, biomaterials and electronics.^[2] Keeping them alive is a large part of the engineering.

Cells need oxygen and nutrients. Waste has to go somewhere. Temperature and chemistry matter. Thick tissue is hard to sustain, and the joint between biological tissue and synthetic structure has to hold without damaging the cells. Running outside a controlled laboratory remains difficult.^[3]

So a robot with living actuators needs more than a power supply. It needs metabolic support. Maintenance starts to look like tissue care.

A robot that exercises

In 2026, researchers demonstrated OstraBot, a muscle-powered swimmer. During development, pairs of engineered skeletal-muscle tissues were arranged so their spontaneous contractions repeatedly loaded each other. The tissues trained against one another, which strengthened and aligned the muscle before it went into the robot. The swimmer reached up to 15.6 body lengths per minute.^[4]

An electric motor arrives with a specification. Living muscle arrives with a history. Its performance partly reflects what happened to it while it grew.

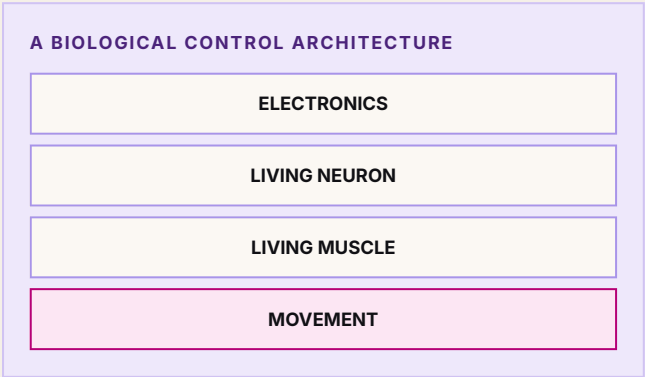
That adds design variables robotics has never had. How should the tissue mature? What stresses should it see? How long should it be conditioned? How do you keep its performance after deployment? Those sound as much like physiology as mechanical engineering.

Give the muscle nerves

Muscle is only one part of biological movement. In animals it sits inside a system of nerves, sensory feedback, connective tissue and metabolic regulation. Most biohybrid robots still depend on external stimulation and laboratory support, and closed-loop control, long-term viability and metabolic autonomy remain open problems.^[3]

Soft electronics may help. Flexible electrodes can stimulate tissue while deforming with it, and soft sensors can track force and strain without the mechanical mismatch of rigid parts.^[5]

In 2024, a bioelectronic neuromuscular robot went a step further. It contained human-derived motor neurons and cardiac muscle cells. Electronics stimulated selected neurons, the neurons activated the muscle, and the muscle moved robotic fins.^[6]



That is a biological control architecture inside a machine. Robotics, tissue engineering and neuroengineering start to overlap.

Start from the animal instead

Cyborg-insect research takes the opposite route: begin with a whole organism. In 2025, a team coordinated Madagascar hissing cockroaches carrying miniature electronic controllers. Electrical stimulation influenced their movement while algorithms steered the group through obstructed terrain.^[7]

From an engineering view, it is obvious why. Evolution already supplied the legs, joints, balance, terrain adaptation and energy efficiency. The control layer is what gets added.

The organism itself becomes part of the robotic platform. That moves the ethical boundary, because the capabilities engineers want belong to an animal. Welfare stops being a separate philosophical topic and becomes a design constraint. I expect the same question to follow every living component we build: what do we owe it?

Remove the artificial body

Xenobots go further still. In 2020, researchers used evolutionary computation to explore candidate body configurations, then manually built selected designs from frog cells. The cellular structures could move and show simple collective behaviour.^[8] There was no conventional robotic substrate. The cells formed the whole body, and AI contributed earlier, by searching for useful morphologies.

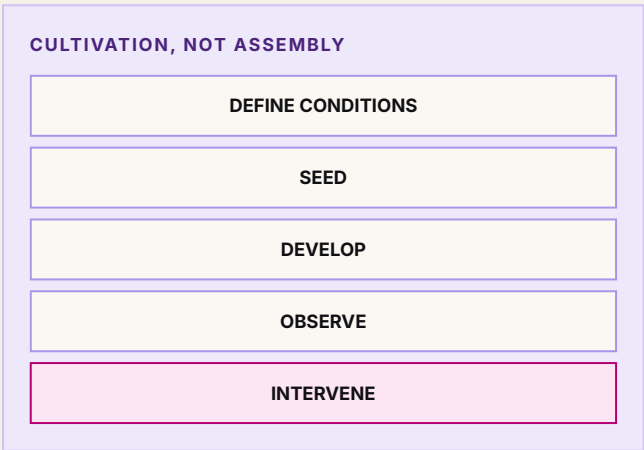
The intelligence doesn't have to sit inside the machine. It can work upstream, shaping a form that biological material later realises.

The work drew far more attention when researchers demonstrated kinematic self-replication. Some constructs could gather loose frog cells into clusters that, under suitable laboratory conditions, developed into new motile constructs. Computational search helped find body shapes that extended the process.^[9] It depended on supplied cellular material in a prepared environment. There was no indefinite autonomous reproduction. The result is remarkable enough without the science-fiction gloss.

Fabrication starts to look like cultivation

Anthrobots show it best. Researchers began with adult human airway cells. Under specific culture conditions, individual cells multiplied and self-organised into motile multicellular structures covered in cilia. No genetic modification. No inorganic scaffold prescribing the final geometry. Much of the form came from the cells themselves.^[10]

That changes the workflow. Traditional engineering runs from design to manufacture to assembly. Working with living material looks more like this:



The closest analogy is gardening. A gardener doesn't fabricate a tree, yet still makes deliberate choices about conditions, constraints and intervention. The material has developmental competencies of its own, and engineering becomes partly about steering them.

A machine with a life history

A 2025 follow-up study of Anthrobots followed their formation, maturation, limited self-repair and eventual death, and found extensive changes in gene expression as adult airway cells reorganised into the new form.^[11] Morphogenesis, maturation, repair, development, death: not words we usually apply to machines.

A manufactured machine ideally leaves the factory finished. A living system may still be becoming what it is.

In 2026, researchers added a nervous system. Neurobots are living multicellular constructs containing neural precursor cells that differentiated into neurons, extended processes and formed active networks. Compared with non-neural constructs they developed different shapes and movements, and responded differ-

ently to a neuroactive drug.^[12] Those neurons were organising inside a body plan evolution had never met, and they still connected and influenced behaviour. That makes Neurobots interesting to developmental biology as much as to robotics.

Current experiments do not show consciousness or subjective experience. Neural activity and cognition are different claims, and the distinction matters.

The role of the engineer changes

Line the examples up and a sequence appears: use biology as a material, organise it together with synthetic parts, appropriate an existing organism, construct new cellular architectures, cultivate self-organising cells, and let specialised tissue such as neurons develop inside novel bodies.

Along that sequence, less of the final outcome is specified directly. Engineering shifts toward constraints, conditions and feedback. That demands a different kind of control.

THE ENGINEER'S ROLE SHIFTS

SPECIFY

components & geometry

CONSTRAIN

conditions & interfaces

CULTIVATE

development & feedback

And where is the AI?

Often, not inside. Muscle-powered robots can run on simple external stimulation. Anthrobots self-organise through cellular processes. Neurobots carry biological, not artificial, neural networks.

The link to Physical AI is in how these capabilities might combine. AI is good at search, prediction, control, coordination and design. Biology brings growth, self-assembly, actuation, adaptation and repair. Xenobots are one early example: computational search found the morphologies, and cells supplied the physical

system. Cyborg swarms are another: algorithms coordinate while living insects provide locomotion and adaptation.

Autonomy is the hard part

Today's biohybrid machines still lean heavily on laboratory infrastructure. They need controlled environments and external stimulation. Nutrient delivery and waste removal are hard. Long-term tissue-machine interfaces remain a challenge. Metabolic autonomy is largely absent, and closed-loop sensing and control are only starting to mature.^[3]

So the field has many impressive components and few truly autonomous integrated systems. A self-maintaining biohybrid that could sense, adapt, repair and sustain its own living parts would be something qualitatively different. That remains future work.

Where the machine ends

Robotics has traditionally used materials whose behaviour engineers try to specify completely. Living material brings its own dynamics into the design. It contracts, grows, repairs, changes with experience and organises collectively.

Some components will be manufactured. Others will be grown. Some tissues will be trained, others left to self-organise. AI may search the design space, electronics may supply precision, and biology may contribute adaptation, development and repair.

These systems will fit neither the categories we inherited from robotics nor those from biology. That may be the most useful thing about them. The boundary between machine and organism gets less helpful once the engineering itself starts crossing it.

If a machine you depend on grows, heals and eventually dies, is it still a product, or something you look after?

Next: the same boundary from the other side, where the machine connects to a person's own nervous system.

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